

Big Data Application Development

Analytics project

Analytics project

Requirements

Each team has 2–3 members.

- One- or four-member teams are acceptable with the approval of the professor.

At least one data source per team member.

- If there are two team members, there must be at least 2 data sources used in your analytics.

Use Spark for data profiling, cleaning, and all analytics.

- All team members must use Spark to profile and clean their respective data sources.
- Then, the whole team should work together and use Spark (including Spark SQL, Spark Streaming or Structured Streaming, MLlib, GraphX/GraphFrames) to perform data analytics.

Technologies that don't scale cannot be used.

Analytics project

Data profiling and cleaning

Data profiling

Write Spark code to **characterize (profile)** the data in each **column** you plan to use from the data source you are responsible for.

- You can use either the low-level **RDD** interface or the high-level **DataFrames** and **Datasets**.

Data cleaning

Write Spark code to **ETL (extract, transform, load)** your data sources as needed.

- You might want to **drop** some columns.
- You might want to **normalize** data in a column (e.g., “New York City”, “NYC”, “nyc”).
- You might want to **detect badly formatted rows** that might be missing important data.

Analytics project

Timeline

Milestones	Check-in date	Grading
Team forming and project proposal (Brightspace — More Tools — Groups)	Friday, April 3	1% of the final grade (one submission per team)
Data profiling and cleaning	Friday, April 24	4% of the final grade (individual submissions)
Presentation	Tuesday, May 5 (in-class symposium)	5% of the final grade (one submission per team)
Final code and report	Thursday, May 7 (hard deadline, no extensions)	10% of the final grade (one submission per team)

Analytics project

Project proposal

An **analytics** provides **actionable insights** through the analysis of one or more datasets.

Think about a problem you'd like to solve or study.

Read the New York Times article (NYU has free access):

- Eight (No, Nine!) Problems With Big Data, by Gary Marcus and Ernest Davis.

If you did the **Realtime and Big Data Analytics** project before, you are allowed to use the same data sources, but **you must do everything in Spark** now.

Analytics project

Data sources

- How can you obtain a copy of the data? Who owns the data? Is it public or private?
- How much data is there in each source? Is it MB? GB? TB?
- If the data is very large, where can you store it?
- How do you gain access to the data? How long will it take to get access/be approved?
- Who must you ask for permission to access the data?
- Is the data static, periodic, or near realtime?
 - If it's static, you will copy it once, and the source data will not grow.
 - If it's periodic, you will copy it periodically, and your files will be growing over time.
 - If it's near realtime, you will be continually reading in the data.
- If the data is being collected in near realtime, what is the velocity of the data and the volume per unit of time? Can your Spark environment support this?

Analytics project

Brainstorming

Computing

- Internet of Things (IoT)
- Robotics
- Linked data
- Security

Analytics project

Brainstorming

Robot example (IBM research)

- Robot collects temperatures, humidity, etc. at various elevations in the datacenter.
- Based on Roomba system, modifications to make perforated tiles visitable.
- Use realtime data and historical big data to generate on-demand heat maps of datacenter.
- Facilitates diagnosis of hot and cold spots in a datacenter.
- Supports software that suggests actionable insights for alleviating hot and cold spots.

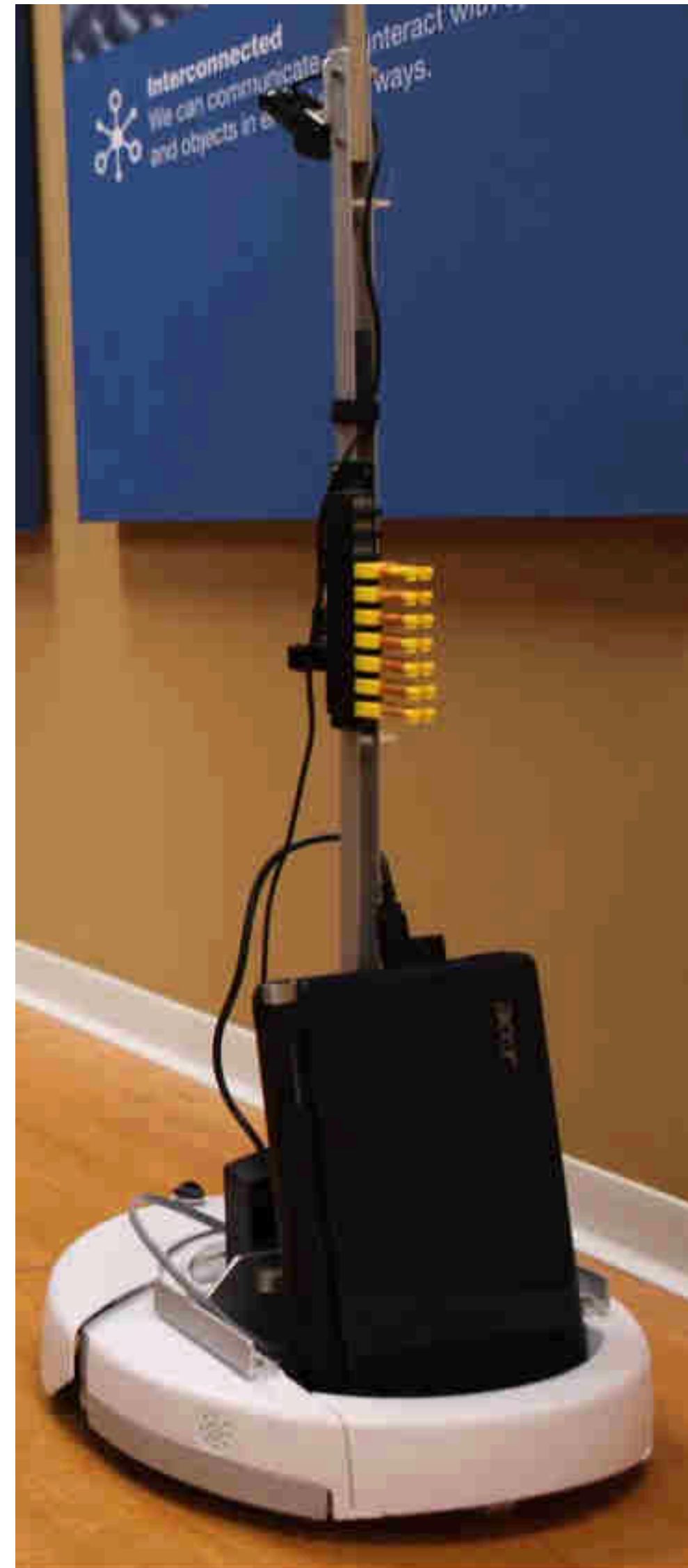


2-meter pole
arrayed with sensors

Webcam

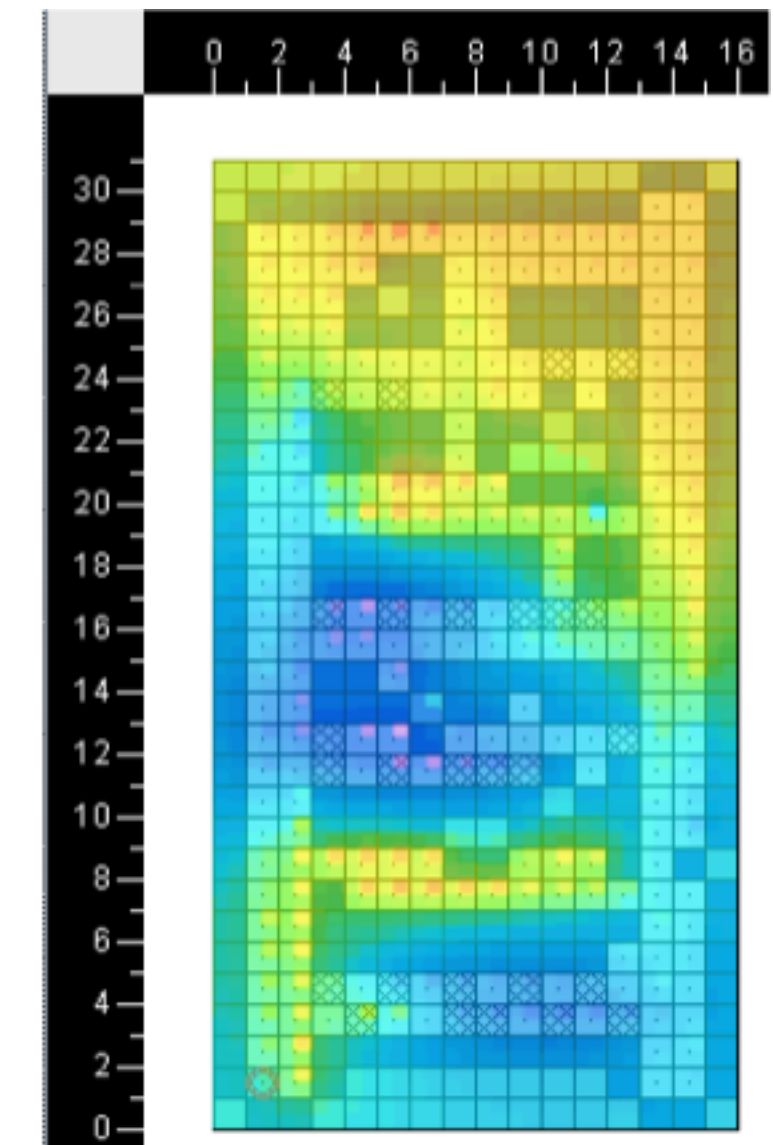
Netbook

Roomba base



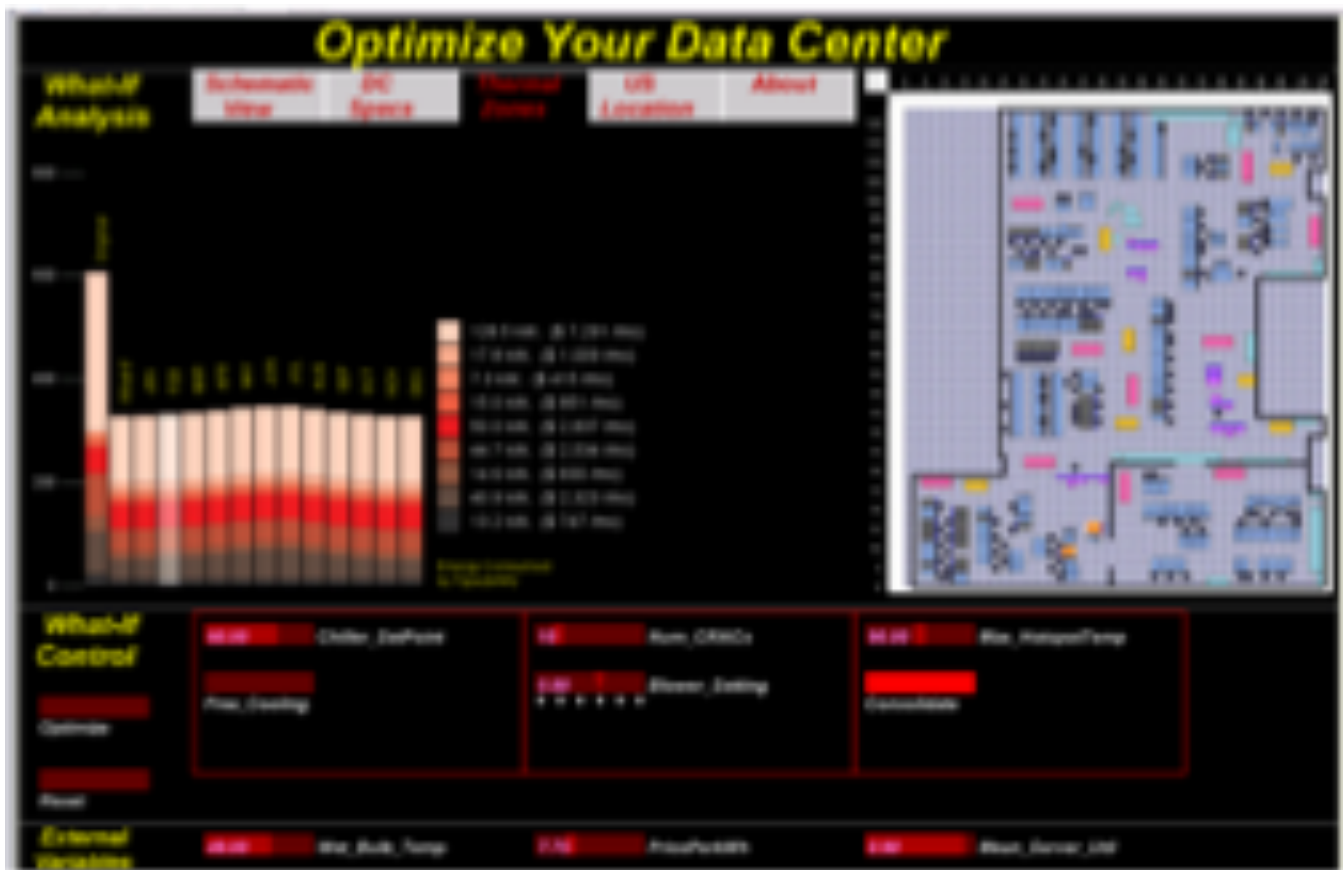
No prior knowledge of data center is needed.
Just set it down and turn it on.

Uses square tile grid for navigation
and full data center coverage.



Asset layout and temperature map
collected by robot at Poughkeepsie
Green Data Center.

Data are used to generate physics and asset models on which diagnostic and predictive analytics rely.



Predictive energy analytics assists diagnosis and can support preview of remedial actions.

Merges IT data with facilities data.

Identifies most likely causes of hot spots; estimates temperature impact and confidence level.

Hot Spot Diagnosis

Green Innovation Data Center - High Density Room

The temperature is 4.8 degrees C above normal at location 22.0, 30.0, 3.5

[Explore
interactively with
HC Dashboard...](#)

	Causes	Impact (Degrees C)	Confidence	Remediation Options																								
	Heavy workload <i>*High workload is contributing to the hot spot.</i> <table><tr><th>Server</th><th>CPU Util Curr (%)</th><th>CPU Util Rise (%)</th></tr><tr><td>vmesxres01.ESX</td><td>99</td><td>84</td></tr><tr><td>vmesxres02.ESX</td><td>100</td><td>91</td></tr><tr><td>vmesxres03.ESX</td><td>99</td><td>94</td></tr><tr><td>vmesxres04.ESX</td><td>99</td><td>94</td></tr><tr><td>vmesxres05.ESX</td><td>96</td><td>84</td></tr><tr><td>vmesxres06.ESX</td><td>99</td><td>87</td></tr><tr><td>vmesxres07.ESX</td><td>100</td><td>95</td></tr></table>	Server	CPU Util Curr (%)	CPU Util Rise (%)	vmesxres01.ESX	99	84	vmesxres02.ESX	100	91	vmesxres03.ESX	99	94	vmesxres04.ESX	99	94	vmesxres05.ESX	96	84	vmesxres06.ESX	99	87	vmesxres07.ESX	100	95	0.4	High	Evacuate workload from these servers...
Server	CPU Util Curr (%)	CPU Util Rise (%)																										
vmesxres01.ESX	99	84																										
vmesxres02.ESX	100	91																										
vmesxres03.ESX	99	94																										
vmesxres04.ESX	99	94																										
vmesxres05.ESX	96	84																										
vmesxres06.ESX	99	87																										
vmesxres07.ESX	100	95																										
	Perforated tile covered <i>*Perforated tile(s) may be blocked.</i>	1.6	Medium	Send data center robot to check area... 																								

Analytics project

Brainstorming

Smarter solutions

- Smarter buildings
- Smarter infrastructure
- Smarter transportation
- Smarter security
- Smarter datacenter

Analytics project

Brainstorming

Smarter conference room example

- Conference rooms at a premium.
- Often, rooms are reserved but go unused.
- Reservation system shows room as reserved even though unoccupied.
- Solution: CO₂ sensors as presence detectors.
- If the room was reserved but is not in use, make the room available for reservation by someone else.

Analytics project

Brainstorming

Politics

- Election
- Hot topics
- Peace

Analytics project

Brainstorming

Election night example

- Tweets are available from the election night.
- Could the winner have been predicted from tweet analysis? At what point?
- Compare analysis of tweets to exit poll results — do they track?
- Can the next election be better called through tweet analysis than exit polls?

Analytics project

Brainstorming

Community service

- Meals on Wheels
- World Health Organization
- Personal safety
- Jobs
- UN Global Pulse

Analytics project

Brainstorming

Sustainable energy

- NYU Green Grants (up to \$20,000 award)
- Recycling
- Renewables
- Solar energy
- Portable drinking water

Alternative project

For most students, doing the **analytics project** is recommended.

Please talk to me if you have done an analytics project in the RBDA course and want to proceed with an **alternative project**.

- Write the connector for a third-party database.
- Write a Spark library for an analytics algorithm.
- ...